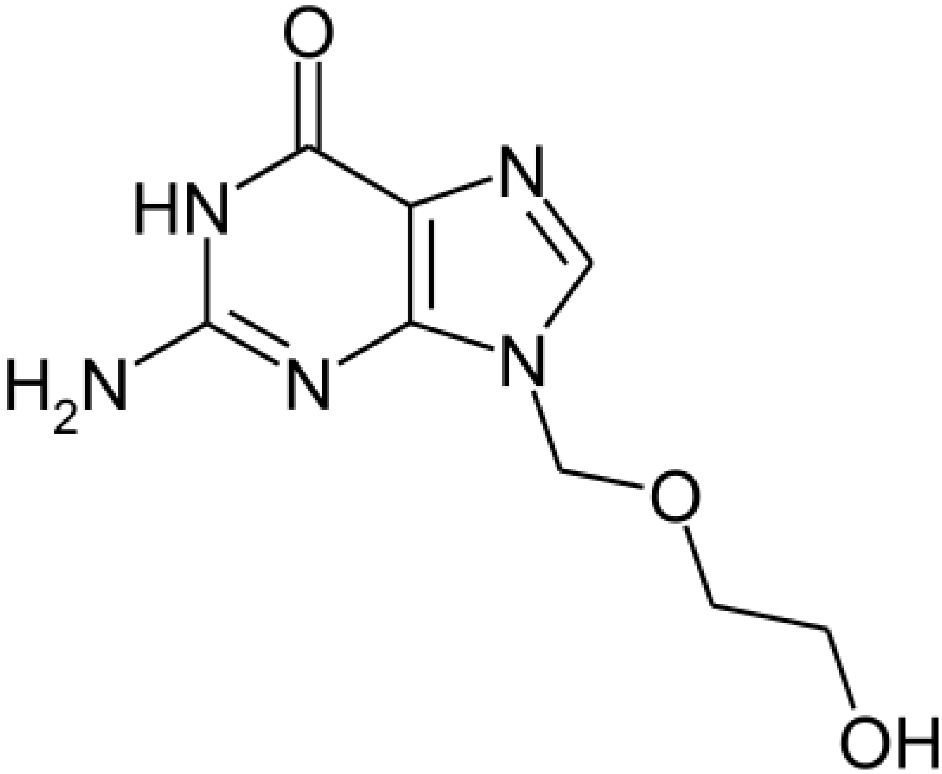

Aciclovir Dynamics:

A Mathematical Model

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Summary

In this report, the development process of a model capable of describing the dynamics of the antiviral drug aciclovir throughout the gastrointestinal tract and the blood is presented. The model is composed of differential equations and the simulations of such model were performed using Euler's method. The model proposed resembles the real behavior of the phenomena; however, to obtain realistic numerical results further considerations, which are discussed, have to be taken into account.

Contents

List of variables

t: time.
$M_{GI}(t)$: Aciclovir mass present in the contents of the gastrointestinal tract at time t.
$M_{blood}(t)$: Aciclovir mass present in the blood at time t.

Chapter 1

Introduction

Aciclovir is an antiviral drug used to combat the members of the herpesvirus family. The intention of this report is to present the development process of a model capable of describing the dynamics of aciclovir throughout the body; namely, how aciclovir taken orally in the form of pills is diffused from the gastrointestinal tract to the bloodstream, and how this drug is then eliminated from the body. This model could aid doctors in prescribing the correct dosage of aciclovir for a given patient. The model in its current state does not account for three important factors that affect the diffusion of medicine from the gastrointestinal tract to the blood. The fist one is a variable volume of gastrointestinal tract contents; which is mostly altered by ingestion. The second and last is a variable clearance with respect to the drug's blood concentration; the amount of blood that is cleared from the blood per unit time. The third one is the bioavailability; the percentage of medicine which trespasses the gastrointestinal tract barrier and reaches the bloodstream. The inclusion of bioavailability into the model is specially important for the case of aciclovir since it is reported to lie between the low values of 10% and 20%. [1] Chapter number two deals with the first modeling cycle, in which the change in the mass of aciclovir in the blood with respect to time was modeled by a single differential equation assuming the concentration of aciclovir in the contents of the gastrointestinal tract to be constant. A simulation was then performed using Euler's method. In the third chapter this concentration was allowed to vary with time, thus incorporating a second differential equation. Coupled with that, the intake of a single pill, as well as the intake of several were simulated, again, using Euler's method.

Chapter 2

Dynamics of aciclovir considering a constant concentration within gastrointestinal tract contents

The purpose of this chapter is to present a first model that describes how aciclovir is transported from the gastrointestinal tract to the blood and then eliminated by making a gross assumption: that the concentration of aciclovir within the gastrointestinal tract contents is constant. This first model is very far from accurately describing the situation presented in the introduction. The model does not take into account the consumption of discrete quantities of medicine in the form of pills, but rather a continuos intake such that the concentration of aciclovir in the contents of the gastrointestinal tract remains constant. Section 2.1 introduces the differential equations that model the situation and explains the meaning of each variable and constant in it. Section 2.2 deals with the selection of appropriate values for the constants in the model. Finally, in section 2.3, the results of the simulation using Euler's method are presented.

2.1 The differential equation

$$\frac{dM_{blood}(t)}{dt} = d_1 \left(C_{GI} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \tag{2.1}$$

$$C_{GI} = \frac{M_{GI}}{V_{GI}}$$

Where t is time, $M_{blood}(t)$ is the aciclovir mass present in the blood at time t, C_{GI} is the concentration of aciclovir in the contents of the gastrointestinal tract, V_{blood} is the blood volume in the body, d_1 is the transfer rate of the drug from the gastrointestinal tract to the blood, d_2 is the clearance of the drug from the blood, M_{GI} is the aciclovir mass present in the gastrointestinal tract and V_{GI} is the volume of gastrointestinal tract contents.

2.2 Selection and determination of appropriate values for the constants

Constants where selected to fit an average adult in a fasting condition. If this model where to be applied to a specific patient, the values of the constants would have to be changed; it is evident that some constant are far more variable from patient to patient while other ones are much more conserved among them.
As pills for treating Herpes Zoster (commonly known as Shingles) generally contain 800 mg of aciclovir, this mass was considered to be present in the gastrointestinal tract (M_{GI}). As for the volume of the gastrointestinal tract contents (V_{GI}), it was reported to be, in average, 0.163 L with a standard deviation of 0.102 L while fasting. [2] An average man has about between 4.5 and 5.5 L of blood (averaging to about 5 L) running trough his body (V_{blood}). [3] The clearance for aciclovir per unit mass was reported to be 51 ml/minmg for patients with a healthy kidney; if an 800 mg pill is taken, then the clearance (d_2) is 0.306 L/h. [4].
The transfer rate of aciclovir from the gastrointestinal tract to the blood (d_1) was not found in literature. This first model does not resemble the real behavior of the phenomena, because of this, the interest was not to obtain values that compare to experimental results. Then, for this first cycle the constant was trivially defined to be 1 L/h.

2.3 Simulation

Euler's method was implemented using Python and the numpy library. A simulation of equation 2.1 with $M_{blood}(0) = 0$ was then run.
The most important thing to realize from figure 2.1 is that $M_{blood}(t)$ reaches an stable equilibrium.

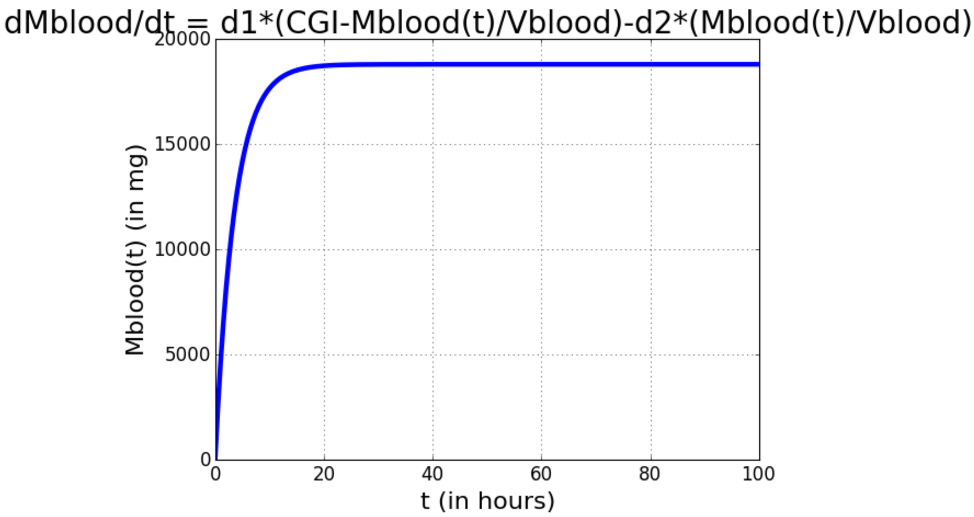

Figure 2.1: **Variation of aciclovir mass in blood with respect to time according to eq. 2.1** The values used for the constants where the ones presented in section 2.1.2

Chapter 3

Dynamics of aciclovir considering a variable concentration within gastrointestinal tract contents

The purpose of this chapter is to present a second model; one that describes how aciclovir is transported from the gastrointestinal tract to the blood and then eliminated considering a variable concentration of aciclovir in the contents of the gastrointestinal tract. Because this second model still does not take into account the bioavailability, the results wont be accurate, but the behavior will be close to that of the real phenomena. Section 3.1 introduces the system of differential equations that model the situation. In section 3.2 an appropriate value for the transfer rate of the drug from the gastrointestinal tract to the blood (d_1) is determined to fit experimental data. In section 3.3, the results of several simulations using Euler's method are presented.

3.1 The system of differential equations

$$\frac{dM_{GI}(t)}{dt} = -d_1 \left(\frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \tag{3.1}$$

$$\frac{dM_{blood}(t)}{dt} = d_1 \left(\frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \tag{3.2}$$

Where $M_{GI}(t)$ is the aciclovir mass present in the gastrointestinal tract at time t and V_{GI} is the volume of the contents of the gastrointestinal tract.

3.2 Determination of the transfer rate of aciclovir from the gastrointestinal tract to the blood

The data recovered by P. Susantakumar, *et al.* for patients who were given 800 mg of aciclovir, shows that the mean peak aciclovir concentration of the drug in the blood was reached at about 2 h after the dosage. [5] When a value of 0.2 L/h for the transfer rate of aciclovir from the gastrointestinal tract to the blood (d_1) was used, the model indeed reached the peak aciclovir concentration at about two hours after the dosage.

3.3 Simulations

A simulation of the system composed of equations 3.1 and 3.2 with $M_{GI}(0) = 800mg$ and $M_{blood}(0) = 0$ was run; the result is shown in figure 3.1.
It is usually indicated for adults who suffer a shingles episode to take pills that contain 800 mg of aciclovir every 4 h for 7 to 10 days. [6] Figure 3.2 shows the first 24 h for a single take. Figure 3.3 shows a complete treatment consisting of 7 days; the simulation was ran until the tenth day. As time passed, the aciclovir mass in the blood started oscillating with respect to a fixed point of approximately 3219.5 mg with a deviation of around ±195.5. When the dosage was cut at the seventh day, the process of completely clearing the aciclovir from the blood was started. After a while, both the aciclovir mass in the blood and in the contents of the gastrointestinal tract started tending more and more to zero. Therefore 0 is a stable equilibrium point for the aciclovir mass in the blood and in the contents of the gastrointestinal tract.

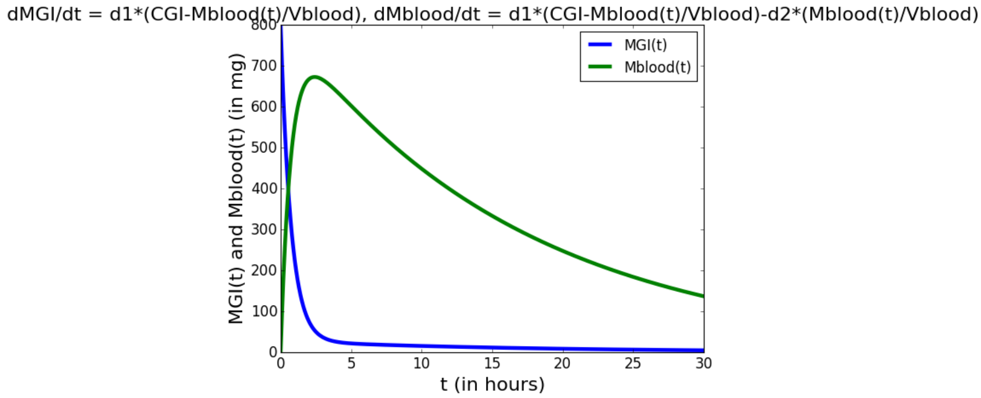

Figure 3.1: **Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 3.1 and 3.2** Single dosage

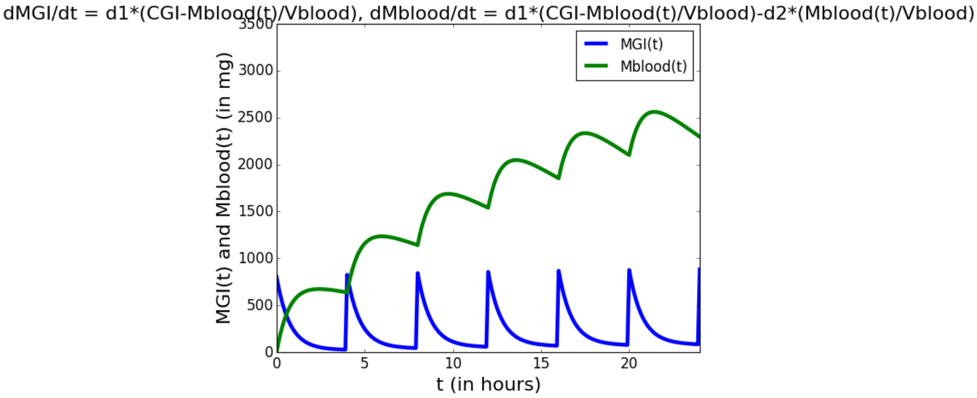

Figure 3.2: **Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 3.1 and 3.2** First 24 h of a usual 7 day treatment

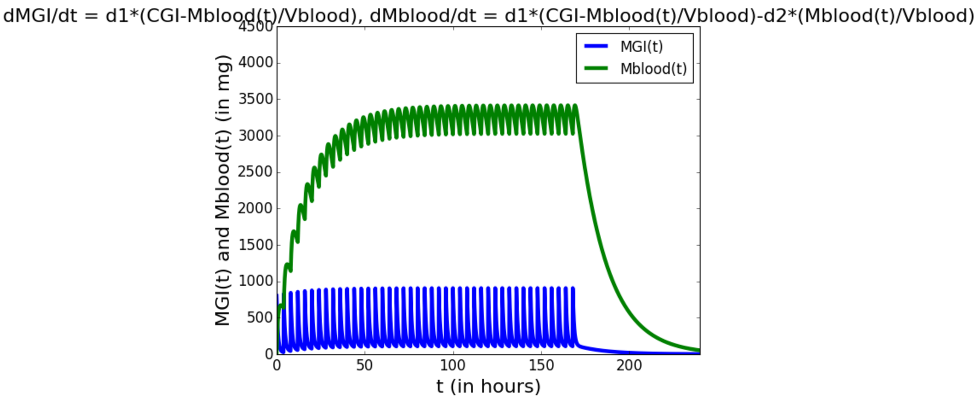

Figure 3.3: **Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 3.1 and 3.2** Complete usual 7 day treatment and the three of following days

Chapter 4

Conclusions

Although the behavior of the model produced during the second cycle (chapter 3) is very similar to the behavior of the real phenomena, more elements have to be incorporated in further modeling cycles to make accurate numerical predictions. Three clear elements to incorporate are a variable volume of the contents of the gastrointestinal tract, a variable clearance and the bioavailability.

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